



Modelling Financial Markets

Seminar 0: Financial Markets in 2026 — Hands-on



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Acronyms used in this material

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- BET: Bucharest Exchange Trading (index)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- CAGR: Compound Annual Growth Rate
- CAR: Cumulative Abnormal Return
- ETF: Exchange-Traded Fund
- FRED: Federal Reserve Economic Data
- FX: Foreign eXchange
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- MAD: Median Absolute Deviation
- MDD: Maximum DrawDown
- OLS: Ordinary Least Squares
- SE: Standard Error
- SR: Sharpe Ratio
- SSR: Sum of Squared Residuals
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)
- VR: Variance Ratio

Today's Question and Required Outputs

- Question of the seminar: **how reliable are the numbers of the Chapter 0 lecture?**
- Route
 - ▶ A1–A4: returns, drawdown, annualisation and CAGR on paper
 - ▶ B6: clean a price file before using it; B1: build a cross-asset table; B3: align two calendars
 - ▶ B1 Extended: attach a standard error and a bootstrap interval to a Sharpe ratio
 - ▶ C2: find the errors in an AI answer; B2 at home
- Required outputs (checklist on the last core slide): the A answers, three figures (B6, B1, B3) and one table per B exercise, the corrected C2 code, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1–A4	returns, drawdown, annualisation, CAGR	Solved	–
B6	which EUR/RON quotes are errors?	Solved	–
B1	how did six asset classes do since 2015?	Solved	–
B3	how do calendars change a correlation?	Solved	–
B1 Extended	how precise is a Sharpe ratio?	Solved	–
C2	what is wrong in an AI answer?	Proposed	A3–A4, B1
B2	how deep and long are drawdowns?	Proposed	A2, B1
Extension packs	A5–A9, B2–B8 extended, B4, B5, B7, B8, C1, C3 (optional, after the seminar)	both	named on each task

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: verification and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

Data Manifest

Series	Symbol, field	Currency	Calendar	Return type
S&P 500 with dividends	SPY.US, adjusted close	USD	US trading days	total return
S&P 500 index	GSPC.INDX, close	USD	US trading days	price
Long Treasuries	TLT.US, adjusted close	USD	US trading days	total return
Gold	XAUUSD.FOREX, close	USD	weekdays only	price
Bitcoin	BTC-USD.CC, close	USD	every day, 24:00 UTC	price
Euro Stoxx 50, Nikkei 225	STOXX50E.INDX, N225.INDX, close	EUR, JPY → USD	local trading days	price
EUR/RON	BNR reference rate; EURRON.FOREX (B6 only)	RON per EUR	Romanian working days	–

- Market files: data/market/<SYMBOL>.csv, columns date, open, high, low, close, adjusted_close, volume; sample 2 Jan 2015 – 18 Sep 2026 unless stated
- FRED exchange rates DEXUSEU (USD per EUR) and DEXJPUS (JPY per USD); BNR EUR/RON; DefiLlama stablecoin supply (extension pack)

Setup: Run and Verify

```
# notebook section "Setup" defines
# fetch_daily(symbol, source, field, start)
# sources: 'market', 'fred', 'bnr', 'defillama'
spy = fetch_daily('SPY.US', 'market',
                  'adjusted_close',
                  start='2015-01-02')
eur = fetch_daily('EUR', 'bnr') # RON per EUR
usd = fetch_daily('DEXUSEU', 'fred')
print(spy.head(), len(spy))
print(spy.index[0], spy.index[-1])
```

- Expected output: 2 945 prices, 2 Jan 2015 to 18 Sep 2026

date	close	adjusted
2015-01-02	205.43	169.26
2015-01-05	201.76	166.24
2015-01-06	199.82	164.64
2015-01-07	202.31	166.69
2015-01-08	205.90	169.65

- The two columns differ: dividends are in adjusted_close only
- If your count or first value differs, stop and check the symbol, the field and the start date

Notation Used in Every Exercise

- ▣ P_t : price on day t ; simple return $R_t = P_t/P_{t-1} - 1$; log return $r_t = \ln(P_t/P_{t-1})$
- ▣ n : number of return intervals; Y : calendar days between the first and the last price /365.25; $q = n/Y$: observed return intervals per year
- ▣ **CAGR** (compound annual growth rate) $= (P_{\text{end}}/P_{\text{start}})^{1/Y} - 1$; **growth multiple** $= P_{\text{end}}/P_{\text{start}}$
- ▣ **Annualised volatility** $\hat{\sigma}_d\sqrt{q}$, with $\hat{\sigma}_d$ the standard deviation of daily log returns
- ▣ **Sharpe ratio** (risk-free rate 0) $\text{SR} = \sqrt{q} \bar{R}/s_R$, from daily simple returns
- ▣ Two series with different calendars: join the *prices* on common days first, then compute returns
- ▣ One worked line: $P_0 = 100$, $P_1 = 102$: $R_1 = 2.00\%$, $r_1 = \ln 1.02 = 1.98\%$

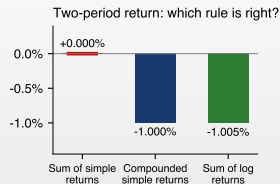
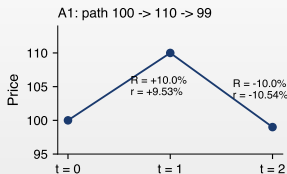
A1 [Solved]: Adding Returns — Task

- ▣ **Question:** can two daily returns simply be added to get the two-day return?
- ▣ **Inputs:** prices $P_0 = 100$, $P_1 = 110$, $P_2 = 99$; no data file
- ▣ **Tasks**
 1. Compute the simple returns R_1, R_2 and the log returns r_1, r_2 .
 2. Add the two simple returns and the two log returns.
 3. Compute the two-period returns $P_2/P_0 - 1$ and $\ln(P_2/P_0)$, and compare them with the sums of step 2.
- ▣ **Report:** a table with columns date, price, simple return, log return, and one sentence on which returns add up
- ▣ Notebook: section A1–A4

A1 [Solved]: Solution

t	P_t	R_t	r_t
0	100	—	—
1	110	+10.00%	+9.53%
2	99	-10.00%	-10.54%
sum		0.00%	-1.005%
0 $\rightarrow$ 2		-1.00%	-1.005%

- Log returns telescope:
 $\ln P_2 - \ln P_1 + \ln P_1 - \ln P_0 = \ln(P_2/P_0)$
- Simple returns compound:
 $(1 + R_1)(1 + R_2) - 1 = 1.1 \times 0.9 - 1 = -1\%$



- Interpretation: adding simple returns reports 0% for a path that lost 1%; add log returns, compound simple returns

A2 [Solved]: Drawdown — Task

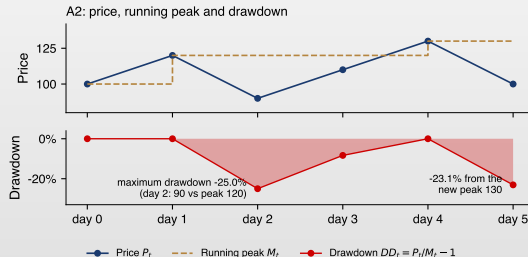
- ▣ **Question:** what is the worst loss from a previous peak, and when did it happen?
- ▣ **Inputs:** prices on days 0–5: 100, 120, 90, 110, 130, 100
- ▣ **Tasks:** complete the worksheet
 1. Running peak $M_t = \max_{s \leq t} P_s$.
 2. Drawdown $DD_t = P_t/M_t - 1$, reported as a negative number.
 3. Maximum drawdown $\min_t DD_t$ and its day.

Day	0	1	2	3	4	5
P_t	100	120	90	110	130	100
M_t	100	120	...	...	...	...
DD_t	0%	0%	...	...	...	...

- ▣ **Report:** the completed table and the maximum drawdown

A2 [Solved]: Solution

Day	0	1	2	3	4	5
P_t	100	120	90	110	130	100
M_t	100	120	120	120	130	130
DD_t	0%	0%	-25.00%	-8.33%	0%	-23.08%



- Maximum drawdown -25% on day 2; the later fall to 100 is smaller, -23.08%, because it is measured from the new peak of 130

A3–A4 [Solved]: Annualisation and CAGR — Task

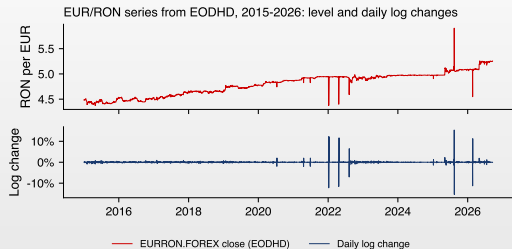
- ▣ **Question:** how do we turn daily numbers into annual ones?
- ▣ **A3 inputs:** daily volatility $\sigma_d = 1\%$; $q = 252$ return intervals a year for an equity index, $q = 365$ for Bitcoin
 1. Compute the annualised volatility $\sigma_d\sqrt{q}$ for both values of q .
 2. State the two assumptions under which $\sqrt{q}$ scaling is exact.
- ▣ **A4 inputs:** S&P 500 price index, multiplied by 3.72 between 2 Jan 2015 and 18 Sep 2026 (4 277 calendar days)
 3. Compute $Y = 4\,277/365.25$ and the $\text{CAGR} = 3.72^{1/Y} - 1$.
 4. Explain why this CAGR is lower than the 13.7% of SPY in B1.
- ▣ **Report:** two volatilities, one CAGR, two sentences

A3–A4 [Solved]: Solution

- A3
 1. $\sqrt{252} = 15.87$, so $1\% \times 15.87 = 15.87\%$; $\sqrt{365} = 19.10$, so 19.10% .
 2. Exact when daily returns are uncorrelated and have constant variance; the extension pack (A6) shows what happens otherwise.
- A4
 3. $Y = 4\,277/365.25 = 11.71$;
 $\text{CAGR} = \exp(\ln 3.72/11.71) - 1 = \exp(1.3137/11.71) - 1 = e^{0.1122} - 1 = 11.87\%$.
 4. The S&P 500 index is a price index: it excludes dividends; SPY's adjusted close reinvests them (13.7% a year).
- Interpretation: CAGR is geometric growth per year, not the average annual return; for the same data the arithmetic mean is higher (extension pack, A5)

B6 [Solved]: Finding Bad Ticks — Task

- **Question:** which EUR/RON quotes in the EODHD series are errors, and how much do they distort volatility?
- **Data:** EURRON.FOREX (close), 1 Jan 2015 – 18 Sep 2026, 3 115 rows; BNR reference rate as the external check
- **Tasks**
 1. Drop weekend rows.
 2. Flag weekday quotes more than 0.05 RON from the 11-day centred rolling median.
 3. For each flag, compare with the BNR rate of that day (or the last earlier fixing on a holiday); remove only quotes more than 0.05 RON from BNR.
 4. Compute annualised volatility of the raw file, the cleaned file and BNR, each with its own q .
- **Report:** an audit table (date, quote, median, BNR, gap, decision) and the three volatilities



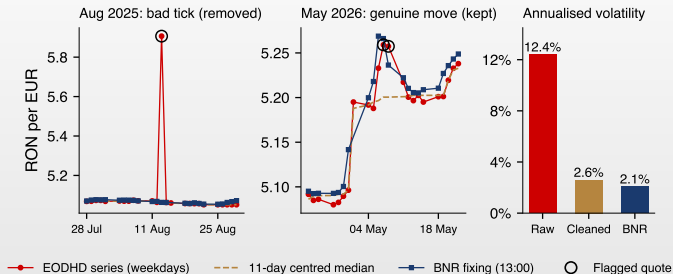
- Look first: the spikes that reverse the next day are candidates
- Notebook: section B6

B6 [Solved]: Solution (1/2) — Audit Table

Date	Quote	Median	BNR (fixing date)	Gap to BNR	Decision
13 Jul 2020	4.7425	4.8368	4.8423	0.100	removed
19 Apr 2021	4.8258	4.9231	4.9261	0.100	removed
28 Jun 2021	4.8265	4.9243	4.9260	0.099	removed
5 Jan 2022	4.3800	4.9435	4.9460	0.566	removed
7 Jan 2022	4.3750	4.9422	4.9436	0.569	removed
22 Apr 2022	4.4000	4.9400	4.9441 (21 Apr)	0.544	removed
9 Aug 2022	4.5900	4.9019	4.9050	0.315	removed
7 Jan 2025	4.9060	4.9738	4.9749 (3 Jan)	0.069	removed
13 Aug 2025	5.9065	5.0636	5.0633	0.843	removed
7 May 2026	5.2591	5.2005	5.2661	0.007	kept
8 May 2026	5.2574	5.2005	5.2364	0.021	kept

- 58 weekend rows dropped; 11 weekday quotes flagged; 9 confirmed errors removed; the May 2026 quotes follow BNR: a genuine move
- Rules: a holiday uses the last earlier BNR fixing; after a deletion the return runs from the previous kept quote to the next one; the centred median uses future quotes, so this is retrospective cleaning, not a real-time filter

B6 [Solved]: Solution (2/2) — Volatility and Interpretation



- Annualised volatility, each with its observed q : raw 12.4% ($q = 266$), cleaned 2.6% ($q = 260$), BNR 2.1% ($q = 251$): nine bad ticks multiply volatility by almost five
- The gap that remains is quote noise: the file quotes another time of day than the 13:00 fixing (daily changes correlate 0.40 with BNR on 2931 common days), lag-1 autocorrelation -0.10 vs $+0.08$ at BNR, 116 file days without a fixing
- Interpretation: use the official BNR rate; a threshold rule removes gross errors, not quote noise

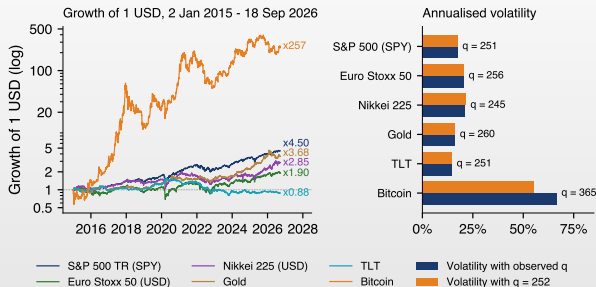
B1 [Solved]: A Cross-Asset Dashboard — Task

- ▣ **Question:** how much did 1 USD grow in six asset classes since 2015, and how volatile was each?
- ▣ **Data:** the six series of the Data Manifest, 2 Jan 2015 – 18 Sep 2026, all in USD
- ▣ **Tasks**
 1. Convert Euro Stoxx 50 and Nikkei 225 to USD: $P^{USD} = P^{EUR} \times (\text{USD per EUR})$,
 $P^{USD} = P^{JPY} / (\text{JPY per USD})$, with the FRED rate of the same day or, if missing, the last rate at most 5 calendar days earlier.
 2. Plot the growth of 1 USD, P_t/P_0 , on a log scale.
 3. For each asset compute n , Y , $q = n/Y$, the multiple, the CAGR and the volatility $\hat{\sigma}_d \sqrt{q}$.
 4. Recompute each volatility with $q = 252$.
- ▣ **Report:** the chart and a table (q , multiple, CAGR, volatility, volatility with 252)
- ▣ **Interpretation:** for which assets does the choice of q matter, and why?

B1 [Solved]: Solution (1/2) — SPY Step by Step

- ▣ SPY (adjusted close): 169.26 on 2 Jan 2015, 761.69 on 18 Sep 2026
 - ▶ multiple $761.69/169.26 = 4.50$; $Y = 11.71$; $\text{CAGR} = e^{\ln 4.50/11.71} - 1 = e^{1.504/11.71} - 1 = 13.7\%$
 - ▶ $n = 2944$ returns, $q = 2944/11.71 = 251.4$; $\hat{\sigma}_d = 1.11\%$, volatility $1.11\% \times \sqrt{251.4} = 17.6\%$
- ▣ Bitcoin: $n = 4277$, $q = 365.25$, $\hat{\sigma}_d = 3.49\%$: volatility 66.7%, but 55.4% with $\sqrt{252}$
- ▣ Gold (weekday quotes): $q = 260.5$, volatility 16.3%, 16.1% with $\sqrt{252}$; the GLD ETF gives 16.3%
- ▣ FX rule in practice: 110 of the Euro Stoxx 50 dates and 104 of the Nikkei 225 dates use an earlier FRED rate (US holidays)
- ▣ The same code runs for every asset: only the symbol, the field and the conversion change

B1 [Solved]: Solution (2/2) — Chart and Table



	q	$\times$	CAGR	Vol.
SPY	251	4.50	13.7%	17.6%
Euro Stoxx 50	256	1.90	5.6%	20.7%
Nikkei 225	245	2.85	9.4%	21.3%
Gold	260	3.68	11.8%	16.3%
TLT	251	0.88	-1.0%	14.7%
Bitcoin	365	256.8	60.6%	66.7%

- q matters for Bitcoin (11 pp), not for gold (0.2 pp)
- Euro Stoxx 50 and Nikkei 225 exclude dividends: their gap to SPY overstates the difference in total return

B3: Why Join Prices before Computing Returns

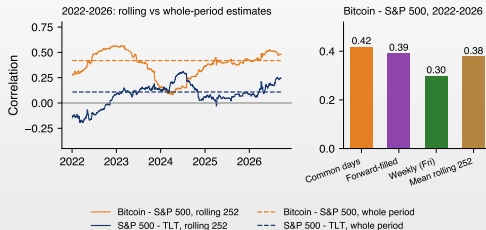
Date	Day	S&P 500 close	Bitcoin close (24:00 UTC)
8 Mar 2024	Friday	5 123.69	68 300.10
9 Mar 2024	Saturday	–	68 498.88
10 Mar 2024	Sunday	–	69 019.79
11 Mar 2024	Monday	5 117.94	72 123.90

- Join prices on common days, then take returns: Friday $\rightarrow$ Monday for both
 - ▶ S&P 500 $\ln(5\,117.94/5\,123.69) = -0.11\%$; Bitcoin $\ln(72\,123.90/68\,300.10) = +5.45\%$
- Returns first, join later: the S&P 500 Friday $\rightarrow$ Monday return (-0.11%) is paired with Bitcoin's Sunday $\rightarrow$ Monday return ($+4.40\%$) and Bitcoin's weekend move is lost
- Forward-filling the S&P 500 on weekends creates two zero returns paired with Bitcoin's weekend moves

B3 [Solved]: Correlations and Calendars — Task

- ▣ **Question:** how much does the calendar and the estimator change a correlation?
- ▣ **Data:** GSPC.INDX, BTC-USD.CC (close) and TLT.US (adjusted close), daily log returns; Bitcoin – S&P 500 in 1 Jan 2022 – 18 Sep 2026; S&P 500 – TLT in 2010–2020 and 2022–2026
- ▣ **Tasks**
 1. Bitcoin – S&P 500 whole-period correlation on common days (prices joined first).
 2. The same with the S&P 500 forward-filled on weekends, and with weekly returns: last common-day price of each week ending on Friday (Thursday if Friday is a holiday); an incomplete final week is dropped.
 3. The mean of the 252-day rolling correlation over 2022–2026 (windows may start in 2021).
 4. S&P 500 – TLT in both periods, with both estimators.
- ▣ **Report:** a table method / sample size / correlation and a plot of the rolling correlations
- ▣ **Interpretation:** which estimator answers “what was the correlation in 2022–2026”?

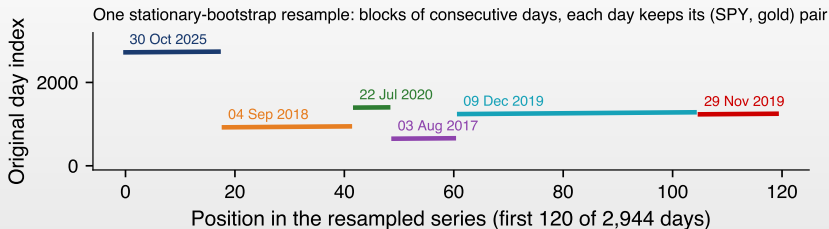
B3 [Solved]: Solution



Bitcoin – S&P 500	n	$\hat{\rho}$	S&P 500 – TLT	2010–20	2022–26
common days	1 182	0.42	whole period	−0.47	+0.11
forward-filled	1 722	0.39	mean rolling 252	−0.43	+0.08
weekly (Friday)	246	0.30	n	2 769	1 182
mean rolling 252	1 182	0.38			

- The whole-period correlation answers the question; the mean of rolling correlations averages one-year windows (some starting in 2021): a different estimand; the calendar alone moves the Bitcoin estimate between 0.30 and 0.42

The Stationary Bootstrap: One Resample



- **Stationary bootstrap** (Politis & Romano (1994)): draw a random start day; continue the block with probability $1 - 1/b$ or jump to a new random day with probability $1/b$ (mean block $b = 20$), wrapping around, until T days are drawn
- Each drawn day keeps the whole vector of returns of that day: the SPY and gold returns stay paired, so their correlation survives
- The resample above has 165 blocks; the first ones start on 30 Oct 2025 (18 days), 4 Sep 2018 (24 days) and 22 Jul 2020 (7 days)
- Recompute the statistic on each resample with q and Y held fixed; 2 000 resamples, generator `numpy.random.default_rng(42)`

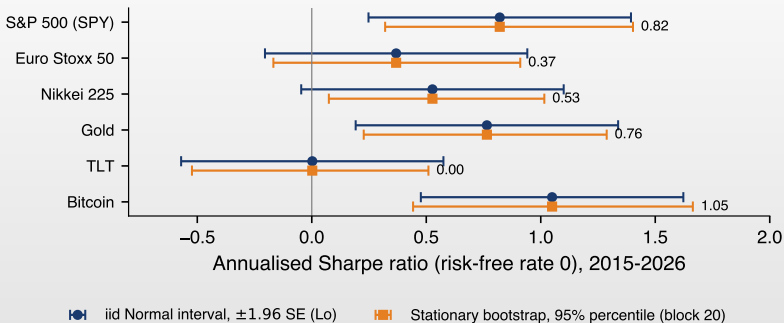
B1 Extended [Solved]: How Precise Is a Sharpe Ratio? — Task

- ▣ **Question:** is SPY's Sharpe ratio precise enough to rank it above gold?
- ▣ **Data:** B1 series, daily simple returns R_t , risk-free rate 0
- ▣ **Tasks**
 1. Compute $SR_a = \sqrt{q} \bar{R} / s_R$ for SPY and the daily $SR_d = \bar{R} / s_R$.
 2. Compute the iid standard error for returns following the Normal distribution (Lo (2002)):
 $SE = \sqrt{q(1 + SR_d^2/2)/T}$, and the 95% interval $SR_a \pm 1.96 SE$.
 3. Stationary bootstrap (block 20, 2000 resamples): 95% percentile interval of SPY's SR; repeat for the other five assets.
 4. On the common days of SPY and gold, bootstrap the paired difference of their Sharpe ratios (both with $q = 252$) and report its 95% interval.
- ▣ **Report:** the SPY calculation, a chart of the intervals, the difference with its interval
- ▣ **Interpretation:** does the table of B1 rank SPY above gold?

B1 Extended [Solved]: Solution (1/3) — SPY Step by Step

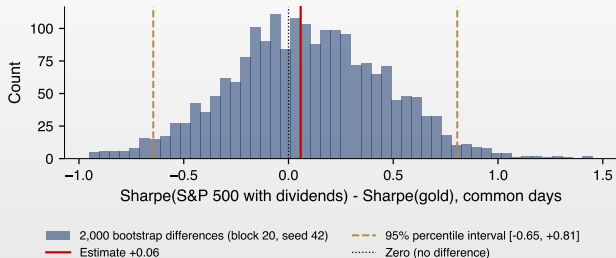
- ▣ Step 1: $T = 2\,944$, $q = 251.4$, $SR_a = 0.82$, so $SR_d = 0.82/\sqrt{251.4} = 0.052$
- ▣ Step 2: $SE = \sqrt{251.4 \times (1 + 0.052^2/2)/2\,944} = \sqrt{0.0855} = 0.29$
 - ▶ this is $\approx 1/\sqrt{Y} = 1/\sqrt{11.71}$: the span, not the frequency, sets the precision
 - ▶ 95% interval $0.82 \pm 1.96 \times 0.29 = [0.25; 1.39]$
- ▣ Step 3: stationary bootstrap, block 20: $[0.32; 1.40]$; the CAGR interval is $[4.4\%; 23.7\%]$
- ▣ Step 4, extension: blocks of 5 and 60 days give $[0.29; 1.39]$ and $[0.37; 1.35]$; the skewness- and kurtosis-corrected SE ([Opdyke \(2007\)](#)) is 0.30
- ▣ Weak serial dependence in daily returns: the three intervals almost coincide

B1 Extended [Solved]: Solution (2/3) — Sharpe Intervals



- Every iid SE is about $1/\sqrt{11.71} = 0.29$: with daily data the SE hardly depends on the asset
- Bitcoin's bootstrap interval [0.44; 1.66] is slightly wider than its iid interval; TLT's contains 0
- The intervals of SPY, gold and Nikkei 225 overlap widely, but overlap is not a test of a difference: next slide

B1 Extended [Solved]: Solution (3/3) — SPY minus Gold



- 2944 common days: difference +0.06, 95% percentile interval [-0.65; +0.81]; 59% of the bootstrap differences are positive
- This is a bootstrap confidence distribution around the estimate, not a distribution simulated under H_0 ; its p-value, twice the smaller tail beyond 0, is 0.82
- Extension: over the 10 pairs of non-crypto assets the smallest p-value is 0.008 (gold minus TLT), above the Bonferroni threshold 0.005
- Interpretation: 11.7 years do not rank SPY above gold; the B1 table describes the sample

C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent to an AI assistant: “Using the daily files in data/market, compute the annualised mean, the annualised volatility and the Sharpe ratio ($r_f = 0$) of SPY and Bitcoin from daily simple returns”
- **Answer:** the code below and the conclusion

```
import numpy as np, pandas as pd
URL = 'https://raw.githubusercontent.com/danpele/MFM/main/data/market/'
def load(sym, field='close'):
    return pd.read_csv(URL + sym + '.csv', index_col='date',
                       parse_dates=True)[field].dropna()
spy, btc = load('SPY.US'), load('BTC-USD.CC')
A = 252                                     # trading days per year
for name, p in [('SPY', spy), ('BTC', btc)]:
    r = p.pct_change().dropna()           # daily simple returns
    mu = r.mean() * A                     # annualised mean
    vol = r.std() * A                     # annualised volatility
    print(name, mu, vol, mu / vol)        # Sharpe ratio, rf = 0
```

- **Conclusion of the AI:** “Sharpe ratios: SPY 0.03, Bitcoin 0.07. Both assets barely reward their risk.”

C2 [Proposed]: Critique an AI Answer — Task

- ▣ **Model:** A3–A4 and B1 [Solved]
- ▣ **Data:** the full files, SPY.US from 29 Jan 1993 and BTC-USD.CC from 13 Jul 2010, both to 18 Sep 2026
- ▣ **Tasks**
 1. Find the three implementation errors planted in the code.
 2. For each, say how you would detect it: a check on the output, a published number, the lecture or the documentation.
 3. Write the corrected code and report the annualised mean, volatility and Sharpe ratio of SPY and Bitcoin.
 4. State one limitation of comparing the two corrected Sharpe ratios, even with the code fixed.
 5. Log every error you actually found, in the format of `AI_ERRORS.md`: request, answer, error, how found, correction.
- ▣ **Interpretation:** does the conclusion “both assets barely reward their risk” survive?
- ▣ AI code usually runs without an error message: the mistakes are in the finance, not in the syntax

B2 [Proposed]: Drawdowns and Recovery Time — Task

- **Model:** A2 and B1 [Solved]
- **Question:** how deep and how long are the drawdowns of the S&P 500 and of Bitcoin?
- **Data:** GSPC.INDX (close) from 3 Jan 2000 and BTC-USD.CC (close) from 17 Sep 2014, both to 18 Sep 2026
- **Conventions:** a spell starts on the last day the running peak was set (equal peaks: the latest) and ends on the first close at or above that peak; a spell still open on 18 Sep 2026 is right-censored
- **Tasks**
 1. Compute DD_t for both assets and plot the two underwater curves.
 2. Report the maximum drawdown with its peak, trough and recovery dates and prices.
 3. Report the longest spell (start, end, calendar and trading days) and the current drawdown.
 4. Repeat 2–3 for the S&P 500 from 17 Sep 2014, the Bitcoin horizon.
- **Interpretation:** why must the two MDDs be compared on the same horizon?

Submission Checklist and Team Repository

Hand in before the next seminar

- ▣ A1–A4: the tables and answers
- ▣ B6, B1, B3: one figure and one table each, as in the solutions
- ▣ B1 Extended: SPY's SE, both intervals, the SPY–gold difference
- ▣ C2: corrected code, table, one limitation, your error log
- ▣ B2: underwater chart and table
- ▣ One interpretation sentence per exercise

Team repository

- ▣ Teams of 3–4; one public GitHub repository, instructors as collaborators
- ▣ README: question, data, how to run
- ▣ First Quantlet folder: B1 Extended with one asset of your choice added
- ▣ Log AI use in AI_USE.md and caught errors in AI_ERRORS.md
- ▣ Extension packs are optional

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Return arithmetic	A5 [Solved], A6, A8	A1–A4, B1 Extended
2. Correlation inference	B3 Extended, A7, A9, C1	B3, B1 Extended
3. Drawdown extremes	B2 Extended	B2, the bootstrap slide
4. Romanian market	B7 [Solved], B8, B5, B5 Extended, B6 Extended	B6, B1, OLS regression
5. Stablecoins	B4, B4 Extended	A3–A4, B7
6. Passive investing lab	C3	the lecture case study, B1 Extended

- Each pack has its own notebook section and starter code; do them in order within a pack
- Useful for the team project: C1 and C3 are possible project starts

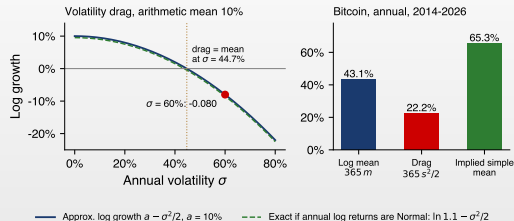
A5 [Solved]: Volatility Drag — Task

- ▣ **Question:** can a positive average return go together with falling wealth?
- ▣ **Notation** (daily): $m = E[r]$, $a = E[R]$, $v_R = \text{Var}(R)$, $v_r = \text{Var}(r)$, with $r = \ln(1 + R)$
- ▣ **Data:** Bitcoin, BTC-USD.CC (close), 17 Sep 2014 (457.33) to 18 Sep 2026 (80 901.46), 4 384 daily returns, 365 a year
- ▣ **Tasks**
 1. Expand $\ln(1 + R)$ to second order and show $m \approx a - (v_R + a^2)/2 \approx a - v_r/2$ for small daily returns.
 2. For Bitcoin, fill a daily-to-annual table: log mean $\times 365$, drag $v_r \times 365/2$, implied simple mean, CAGR $= e^{365m} - 1$; compare with the CAGR from the two endpoint prices.
 3. An asset has an annual arithmetic mean of 10% and a volatility of 60%: approximate its expected log growth and CAGR.
- ▣ **Interpretation:** why can the typical path lose money when the mean return is positive?

A5 [Solved]: Solution

- (a) $\ln(1 + R) = R - R^2/2 + O(R^3)$, so
 $m \approx a - E[R^2]/2 = a - (v_R + a^2)/2$; for small daily returns
 $v_R \approx v_r$ and $a^2 \approx 0$: an approximation, not an identity
- (b) Bitcoin, daily $m = 0.118\%$, $\sqrt{v_r} = 3.49\%$

log mean 365 m	43.1%
drag 365 $v_r/2$	22.2 pp
implied simple mean	65.3%
direct: 365 $\times$ mean of R_t	65.2%
CAGR $e^{0.431} - 1$	53.9%
CAGR from 457.33 to 80 901.46	53.9%



- (c) Approximation: $0.10 - 0.60^2/2 = -0.08$, CAGR $\approx -7.7\%$; if annual log returns follow the Normal distribution, the exact value is $\ln 1.10 - 0.18 = -0.085$, CAGR -8.1%
- The drag of 18 pp exceeds the 10% mean: the typical path loses money

A6 [Proposed]: When the $\sqrt{q}$ Rule Fails — Task

- ▣ **Model:** A3–A4 and A5 [Solved]
- ▣ **Theory** (daily log returns, stationary, variance σ^2 , autocorrelations ρ_j)
 1. For $k = 3$, write $\text{Var}(r_1 + r_2 + r_3)$ as the sum of the nine covariances; then generalise to $\text{Var}(\sum_{t=1}^k r_t) = k\sigma^2[1 + 2\sum_{j=1}^{k-1}(1 - j/k)\rho_j]$.
 2. For a *hypothetical* AR(1), $\rho_j = \rho^j$, find the limit of the bracket as $k \rightarrow \infty$.
- ▣ **Data:** BNR EUR/RON, 4 Jul 2005 – 18 Sep 2026 (`fetch_daily('EUR', 'bnr')`), 5 352 daily log changes
 3. Compute $\hat{\rho}_1$, the $\sqrt{252}$ volatility and its hypothetical AR(1) correction.
 4. Compute the autocorrelations at lags 1–10 and the volatility from month-end changes $\times \sqrt{12}$.
 5. The EODHD series (B6) has $\hat{\rho}_1 = -0.57$: what would the AR(1) factor be, and why is it the wrong fix?
- ▣ **Interpretation:** does positive autocorrelation hide risk or invent it, and is $\hat{\rho}_1$ enough?

A8 [Proposed]: The Standard Error of a Sharpe Ratio — Task

- ▣ **Model:** A5 and B1 Extended [Solved]
- ▣ **Setting:** T iid daily returns with mean μ , variance σ^2 , central moments μ_3, μ_4 (finite), skewness $\gamma_3 = \mu_3/\sigma^3$, ordinary (not excess) kurtosis $\gamma_4 = \mu_4/\sigma^4$; daily SR = μ/σ
- ▣ **Data:** SPY (B1 Extended): $SR_a = 0.82$, $q = 251.4$, $T = 2\,944$, $Y = 11.71$, $\gamma_3 = -0.31$, $\gamma_4 = 17.1$
- ▣ **Tasks**
 1. Gradient: for $g(\mu, \sigma^2) = \mu/\sigma$, compute ∇g .
 2. Quadratic form: $\sqrt{T}(\hat{\mu} - \mu, \hat{\sigma}^2 - \sigma^2) \rightarrow N(0, \Sigma)$, $\Sigma = \begin{pmatrix} \sigma^2 & \mu_3 \\ \mu_3 & \mu_4 - \sigma^4 \end{pmatrix}$; compute $\nabla g^\top \Sigma \nabla g$ and its Normal case (Lo (2002)).
 3. Annual scaling: with $SR_a = \sqrt{q} SR_d$ and $T = qY$, show $\text{Var}(\widehat{SR}_a) = (1 + SR_a^2/(2q))/Y$.
 4. Compute SPY's SE three ways: Normal daily, general daily, and the annual-observation formula $\sqrt{(1 + SR_a^2/2)/Y}$; state the sampling assumption of each.
- ▣ **Interpretation:** why does sampling more often than monthly barely help?

B3 Extended [Proposed]: Has the Stock–Bond Correlation Changed? — Task

- ▣ **Model:** B1 Extended [Solved]
- ▣ **Data:** B3, S&P 500 – TLT daily log returns on common days: 2010–2020 ($n_1 = 2\,769$) and 2022 – 18 Sep 2026 ($n_2 = 1\,182$)
- ▣ **Tasks**
 1. Compute $\hat{\rho}_1, \hat{\rho}_2$ unrounded, then $z_i = \operatorname{atanh} \hat{\rho}_i$, $\operatorname{SE} = \sqrt{1/(n_1 - 3) + 1/(n_2 - 3)}$ and $z = (z_2 - z_1)/\operatorname{SE}$ (Fisher (1915); iid pairs from a bivariate Normal distribution).
 2. Compute the iid 95% interval of $\hat{\rho}_2 - \hat{\rho}_1$: $\pm 1.96 \sqrt{(1 - \hat{\rho}_1^2)^2/(n_1 - 3) + (1 - \hat{\rho}_2^2)^2/(n_2 - 3)}$.
 3. Stationary bootstrap each period separately (block 20, 2 000 resamples) and report the 95% interval of the change.
 4. Compare the widths of the two intervals.
- ▣ **Report:** z , p , both intervals, the width ratio and a histogram of the bootstrap changes
- ▣ **Interpretation:** why is the bootstrap interval wider, and does the test identify the cause of the change?

A7 [Proposed]: Why Is the Weekly Correlation Lower? — Task

▣ **Model:** B3 and B3 Extended

▣ **Data:** $x_t = \text{S\&P 500}$, $y_t = \text{Bitcoin}$, daily log returns on common days, 2022 – 18 Sep 2026, $n = 1182$;
 $\gamma(j) = \text{Cov}(x_t, y_{t-j})$

▣ **Tasks**

1. For sums of 5 consecutive common-day returns X, Y with no own autocorrelation, draw the 5×5 grid of covariances and show $\text{Cov}(X, Y) = 5\gamma(0) + \sum_{j=1}^4 (5-j)[\gamma(j) + \gamma(-j)]$, so
 $\rho_5 = \rho_1 + \sum_{j=1}^4 (5-j)[\gamma(j) + \gamma(-j)] / (5\sigma_x\sigma_y)$.
2. Check the assumption: own lag-1 autocorrelations of x and y .
3. Compute the lag term, the implied ρ_5 and the correlation of overlapping 5-day sums.
4. Compute the weekly correlation for weeks anchored on Monday, ..., Friday, each with a Fisher 95% interval.
5. Report the joint week-block bootstrap interval of daily minus weekly (B3 Extended notebook).

▣ **Interpretation:** is the gap 0.42 vs 0.30 a property of the markets or of the sampling? The [Epps \(1979\)](#) effect predicts the opposite sign

A9 [Proposed]: Asynchronous Closes — Task

- ▣ **Model:** B3 [Solved]
- ▣ **Setting:** log prices $X(s) = W(s) + U(s)$ (Bitcoin) and $Y(s) = W(s) + V(s)$ (S&P 500), W, U, V independent Brownian motions with variance rates $\sigma_W^2, \sigma_U^2, \sigma_V^2$; $\rho = \sigma_W^2 / \sqrt{(\sigma_W^2 + \sigma_U^2)(\sigma_W^2 + \sigma_V^2)}$
- ▣ X closes at integer times t (24:00 UTC); Y closes δ of a day earlier, at $t - \delta$ (20:00 UTC in summer, 21:00 in winter)
- ▣ **Tasks**
 1. Draw the intervals covered by x_t, y_t and y_{t+1} on a time line and show $\text{Corr}(x_t, y_t) = \rho(1 - \delta)$, $\text{Corr}(x_t, y_{t+1}) = \rho\delta$, all other cross-correlations 0.
 2. Deduce that the lag $-1, 0, +1$ correlations sum to ρ (Scholes & Williams (1977)).
 3. Compute δ in summer and winter and the implied correlations for $\rho = 0.45$.
 4. Tabulate the observed $\text{Corr}(x_t, y_{t+j})$, $j = -1, 0, 1$, for 2022–2026 and compare.
- ▣ **Interpretation:** is the asynchronous-close bias important for this pair? Treat the sum as a diagnostic

C1 [Proposed]: Did Spot ETFs Change Bitcoin's Correlation with Equities?

- ▣ **Model:** B3 Extended
- ▣ **Question:** US spot Bitcoin ETFs started trading on 11 Jan 2024; did the Bitcoin – S&P 500 correlation change at that date?
- ▣ **Data:** BTC-USD.CC and GSPC.INDX, daily log returns on common days, 2018 – 18 Sep 2026
- ▣ **Mandatory baseline**
 1. Correlations in 2018–2019, 2020–2021, 1 Jan 2022 – 10 Jan 2024 (pre) and 11 Jan 2024 – 18 Sep 2026 (post), each with a stationary-bootstrap 95% interval (block 20).
 2. Test $H_0 : \rho_{post} - \rho_{pre} = 0$ with a bootstrap interval of the difference; plot the rolling correlation with the ETF date.
- ▣ **Project extensions:** a counterfactual asset without an ETF event (state why it is comparable), an unknown-date break search, other calendars and windows
- ▣ **Report:** a table period / n / $\hat{\rho}$ / interval, the difference, one figure, one limitation; a causal claim needs the extensions

B2 Extended [Proposed]: Is the 2009 Drawdown Extreme? — Task

- ▣ **Model:** B1 Extended [Solved] and B2
- ▣ **Data:** S&P 500 daily log returns r_t , 3 Jan 2000 – 18 Sep 2026, $n = 6\,717$; observed MDD -56.8% , longest spell 1 803 trading days (spells still open at the end are censored at the last day)
- ▣ **Tasks**
 1. Build one path: resample n returns, set $\ln P_t = \ln P_0 + \sum_{s \leq t} r_s^*$, compute its MDD and longest spell; plot it.
 2. Repeat 5 000 times with the iid bootstrap and 5 000 times with the stationary bootstrap (block 20); seed 42.
 3. Plot both distributions of the MDD and of the longest spell; mark the observed values; report medians, 5% quantiles and the tail shares $P(\text{MDD}^* \leq -56.8\%)$ and $P(\text{spell}^* \geq 1\,803)$.
 4. Compute $\hat{\rho}_1$ and $\widehat{\text{VR}}(20)$ from overlapping 20-day returns.
- ▣ **Interpretation:** is -56.8% extreme for this return distribution? The resampled returns include the crisis itself, so this is a diagnostic, not a forecast

B7 [Solved]: Event Study, the Bank Tax of December 2018 — Task

- ▣ **Question:** how much did Banca Transilvania lose on the bank-tax news, beyond the market move?
- ▣ **Event:** tax announced on the evening of 18 Dec 2018; event day 0 = 19 Dec 2018; trading-day index τ
- ▣ **Data:** TLV.R0 (adjusted close) and BET (close), daily log returns; estimation window $[-250, -11]$ (18 Dec 2017 – 4 Dec 2018, $T = 240$); event window $[0, +5]$ ($L = 6$)
- ▣ **Method:** MacKinlay (1997); Campbell, Lo & MacKinlay (1997), Ch. 4; errors iid with variance σ_ε^2 , no change of variance in the event window
- ▣ **Tasks**
 1. Estimate $r_{i,t} = a + b r_{m,t} + \varepsilon_t$ by OLS: report $\hat{a}$, $\hat{b}$, $\hat{\sigma}_\varepsilon$, $\bar{r}_m$ and $S_{mm} = \sum (r_{m,t} - \bar{r}_m)^2$.
 2. Fill a table: day, r_i , r_m , predicted $\hat{a} + \hat{b}r_m$, AR; sum the ARs into $CAR[0, +5]$.
 3. Test $AR_0 = 0$ and $CAR = 0$ with the prediction-error variances $\hat{\sigma}_\varepsilon^2[1 + 1/T + (r_{m,0} - \bar{r}_m)^2/S_{mm}]$ and $\hat{\sigma}_\varepsilon^2[L + L^2/T + (\sum_\tau r_{m,\tau} - L\bar{r}_m)^2/S_{mm}]$, Student t with $T - 2$ degrees of freedom.
- ▣ **Interpretation:** BET contains both big banks: what does this do to $\hat{b}$ and to the AR?

B7 [Solved]: Solution (1/2) — Model and Abnormal Returns

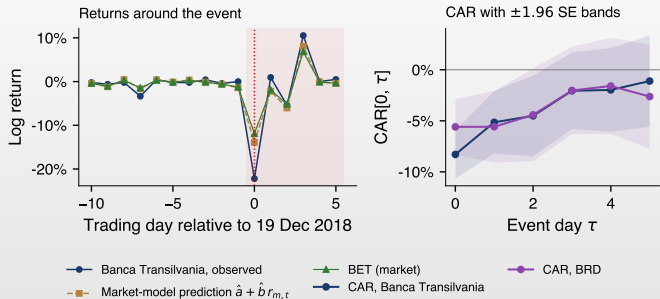
□ Step 1: $\hat{a} = 0.066\%$ a day, $\hat{b} = 1.176$, $\hat{\sigma}_\varepsilon = 0.832\%$, $\bar{r}_m = 0.060\%$, $S_{mm} = 0.01285$

τ	Date	r_i	r_m	$\hat{a} + \hat{b}r_m$	AR
0	19 Dec 2018	-22.21%	-11.89%	-13.91%	-8.29%
1	20 Dec 2018	+0.95%	-1.92%	-2.20%	+3.15%
2	21 Dec 2018	-5.41%	-5.19%	-6.04%	+0.63%
3	24 Dec 2018	+10.54%	+6.82%	+8.08%	+2.46%
4	27 Dec 2018	0.00%	-0.13%	-0.08%	+0.08%
5	28 Dec 2018	+0.50%	-0.37%	-0.37%	+0.87%
CAR[0, +5]					-1.10%

□ Step 3: day 0, $(r_{m,0} - \bar{r}_m)^2 / S_{mm} = 1.112$: $SE = 0.832\% \times \sqrt{1 + 1/240 + 1.112} = 1.21\%$, $t = -6.85$, $p < 0.001$

□ CAR: $SE = 0.832\% \times \sqrt{6 + 0.15 + 1.326} = 2.27\%$, $t = -0.48$, $p = 0.63$: do not reject $CAR = 0$

B7 [Solved]: Solution (2/2) — Chart and Interpretation



- Day 0 is a large surprise, but the ARs of days 1–3 reverse most of it: the six-day CAR is not significant
- BET fell 11.9% on the same news: the market model removes most of the shock, so the AR (−8.3%) is market-relative, not the bank's total loss (−22.2%); a benchmark without banks would avoid this contamination
- The extreme market day more than doubles the variance of AR_0 ; event days may also be more volatile than the estimation window, which even this formula ignores

B8 [Proposed]: The Same Event for BRD — Task

- ▣ **Model:** B7 [Solved]; reuse its function, only the symbol changes
- ▣ **Data:** BRD.R0 (adjusted close) and BET; same windows as B7
- ▣ **Tasks**
 1. Report $\hat{b}$, $\hat{\sigma}_\varepsilon$, the AR table and $CAR[0, +5]$ with t and p .
 2. Build a two-bank table: AR_0 , its prediction-error t , CAR and its t .
 3. Plot the two CAR paths with ± 1.96 SE bands on one chart.
 4. Which bank was hit harder relative to its normal volatility? Use the prediction-error t of AR_0 as the comparison statistic.
- ▣ **Interpretation:** the reversal can mean overreaction or a partly withdrawn policy; what dated evidence would separate the two?

B5 [Proposed]: BET-TR in RON and in EUR — Task

- ▣ **Model:** B1 [Solved] (currency conversion)
- ▣ **Data:** BETTR.INDX (close) and BNR EUR/RON S_t (RON per EUR, `fetch_daily('EUR', 'bnr')`), prices joined on common days, 5 Jan 2015 – 18 Sep 2026
- ▣ **Tasks**
 1. Convert: $P_t^{EUR} = P_t^{RON} / S_t$; work one date step by step (13 Aug 2025).
 2. Show $r_t^{EUR} = r_t^{RON} - \Delta s_t$, with $s_t = \ln S_t$.
 3. Compute the CAGR and the volatility (observed q) in RON and in EUR, and the correlation of r^{RON} with Δs .
 4. Plot both wealth paths.
- ▣ **Report:** the two CAGRs, two volatilities, the correlation, the chart
- ▣ **Interpretation:** for a euro investor, is the leu's depreciation a return drag, a risk, or both?

B5 Extended [Proposed]: Is the Currency Effect Significant? — Task

- ▣ **Model:** B1 Extended and B7 [Solved]
- ▣ **Data:** B5, $n = 2925$ common-day returns, $q = 250.0$ observed intervals a year
- ▣ **Tasks**
 1. Show $\text{Var}(r^{EUR}) = \text{Var}(r^{RON}) + \text{Var}(\Delta s) - 2 \text{Cov}(r^{RON}, \Delta s)$; report each term annualised ($\times q$) in decimal units, then take square roots.
 2. Regress r_t^{RON} on a constant and Δs_t with Newey–West HAC standard errors (Newey & West (1987), $L = 8$); report the slope, t and p ; add a stationary-bootstrap 95% interval for the correlation.
 3. Estimate the annual log drift of the leu, $q \times$ mean of Δs_t , with its HAC SE, t and 95% interval.
 4. Show that $\ln(1 + \text{CAGR}^{RON}) - \ln(1 + \text{CAGR}^{EUR}) = \ln(S_{\text{end}}/S_{\text{start}})/Y$, the drift of (c).
- ▣ **Interpretation:** the hypothesis tested in (c) is “zero log drift”; is the 1.6 pp CAGR gap distinguishable from zero?

B6 Extended [Proposed]: A Calibrated Threshold — Task

- ▣ **Model:** B6 [Solved]
- ▣ **Method:** Hampel filter ([Hampel \(1974\)](#)): flag x_t if $|x_t - m_t| > \max(c \times 1.4826 \times \text{MAD}_t, \text{floor})$; m_t = median of the centred 11-quote window (truncated at the ends), MAD_t = median of $|x_j - m_t|$ in the same window
- ▣ **Data:** BNR EUR/RON 2015–2026 (calibration, 2 939 days) and the weekday EODHD series of B6 (application)
- ▣ **Tasks**
 1. Work the 11-quote window around 13 Aug 2025 step by step: m_t , MAD_t , the threshold for $c = 11$.
 2. On BNR, plot the share of flagged days against c (grid 1, 1.5, ..., 59.5) for floors 0, 0.01, 0.02, 0.03 RON; find the smallest c with at most 0.1% flags.
 3. Apply the calibrated rules (no floor; floor 0.01) to the EODHD series, remove all flagged quotes and recompute the volatility.
- ▣ **Interpretation:** why MAD and 1.4826, and what happens when MAD_t is almost zero?

B4 [Proposed]: Stablecoin Growth — Task

- ▣ **Model:** A3–A4 and B1 [Solved]
- ▣ **Data:** DefiLlama, daily total circulating value of USD-pegged stablecoins, field `totalCirculatingUSD.peggedUSD`, divided by 10^9 (bn USD), 31 Dec 2019 – 18 Sep 2026:
`fetch_daily('stablecoincharts/all', 'defillama')`
- ▣ **Tasks**
 1. Extract year-end values 2019–2025 and the calendar-year growth; report 2026 separately as year-to-date growth to 18 Sep.
 2. Find the 2022 peak and the following trough up to the end of 2023, with dates, and the peak-to-trough change.
 3. Compute the CAGR from 1 Jan 2020 to 18 Sep 2026 with the exact endpoints.
 4. Plot the series with the year-end points.
- ▣ **Interpretation:** supply is a USD value in circulation, not a price; can it fall without any token being burned?

B4 Extended [Proposed]: Did the Terra Collapse Change the Growth Regime? — Task

- ▣ **Model:** B7 [Solved] (OLS regression and its t test)
- ▣ **Data:** $g_t = \Delta \ln(\text{supply}_t)$, 2 Jan 2020 – 18 Sep 2026, $n = 2\,452$; Terra collapse 9 May 2022
- ▣ **Tasks**
 1. Known date (Chow (1960)): $g_t = a + b D_t + e_t$, $D_t = 1\{t \geq 9 \text{ May } 2022\}$, HAC SE (Newey & West (1987), $L = 8$); report a , b (annualised $\times 365$) and t .
 2. Unknown date (Andrews (1993)): the HAC Wald statistic for every date in the central 70% (3 Jan 2021 – 15 Sep 2025); compare its maximum with 8.85, the 5% critical value for this trimming.
 3. Least-squares date: the date minimising the sum of squared residuals; plot the Wald and SSR profiles.
 4. Check the residual variance before and after the date.
- ▣ **Interpretation:** did the slowdown start with Terra? Is the break in the level or in the drift of supply?

C3 [Proposed]: Passive Investing and Demand Elasticity — Task (1/2)

- ▣ **Model:** the lecture case study (Haddad, Huebner & Loualiche (2025)) and B1 Extended
- ▣ **Question:** is the one-third pass-through of the paper robust to who turns passive?
- ▣ **Notation:** stock k , active investor i ; w_{ik} = shares held by i as a fraction of all shares of k ; active share $A_k = \sum_i w_{ik}$; own elasticity $\bar{\mathcal{E}}_{ik}$; aggregate elasticity (footnote 53)

$$\mathcal{E}_{agg,k} = \frac{\sum_i w_{ik} \bar{\mathcal{E}}_{ik}}{1 + \chi A_k}$$

- ▣ **Two-investor warm-up:** $w_1 = 0.4$ with $\bar{\mathcal{E}}_1 = 1$, $w_2 = 0.3$ with $\bar{\mathcal{E}}_2 = 3$, $\chi = 2.97$; 0.1 of the shares turn passive
 1. Compute $\mathcal{E}_{agg}$ before, and after the switch when both reduce in proportion, when investor 2 (elastic) switches, when investor 1 switches.
 2. Compare each $\Delta \log \mathcal{E}_{agg} / \Delta \log A$ with $1/(1 + \chi A) = 0.325$.
- ▣ **Data:** FRED B0GZ1LM563064100Q (ETF corporate equities) over B0GZ1LM893064105Q (all sectors), 2001 Q1 – 2026 Q2
 3. ETF share in 2001 Q1, 2020 Q4, 2026 Q2 and the implied changes with 81% active and pass-through 0.33.

C3 [Proposed]: Passive Investing and Demand Elasticity — Task (2/2)

- ▣ **Simulation** (starter code in the notebook, `numpy.random.default_rng(seed)`)
 4. 2 000 stocks: $A_k \sim U(0.55, 0.85)$; 40 active investors per stock with $w_{ik} = A_k \times$ flat Dirichlet weights; $\bar{\mathcal{E}}_{ik}$ log-Normal with median 1.6 and log standard deviation 0.6; $\chi = 2.97$.
 5. In each stock a fraction $f_k \sim U(0, 0.2)$ of A_k turns passive under three rules: all in proportion; most elastic first; least elastic first (whole holdings in order, the last one partly, until $f_k A_k$ is reached).
 6. Regress $\Delta \log \mathcal{E}_{agg,k}$ on a constant and $\Delta \log A_k$ (Table 4, column 1).
 7. Repeat for seeds 1–10 with the same draws for all rules, and for $\chi = 0$.
- ▣ **Report:** the ETF shares and implied changes; a table of slopes (mean, min, max over seeds) next to the mean of $1/(1 + \chi A_k)$
- ▣ The seed range is simulation variability, not a sampling confidence interval
- ▣ **Interpretation:** why is the slope one-third only for representative switchers, and how should the paper's 11% be read if the most elastic investors turned passive first?

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- ▣ Public data: [FRED](#), [BNR reference rates](#), [DefiLlama](#)
- ▣ Code and data: github.com/danpele/MFM